

Have UK Earnings Kept Pace With the Rising Cost of Living Between 2015 and 2025?

Contents:

Data used - Page 2

Methodology - Page 2

Introduction - Page 3

Results - Page 3

Interpretation - Page 4

Conclusion - Page 5

Bibliography - Page 5

Data Used:

All data was sourced from the Office For National Statistics 2025-2015 through reports including:

- CPIH
- Average Weekly Earnings, Total Pay
- Price Index of Private Rents
- Expenditure category CPIH data

Methodology:

Once the ONS data was collated, I utilised the Pandas Library via Python to filter and clean the data, transforming large sets of monthly data into annual averages between the 2015-2025 period. Using indexes would provide a clear insight into the growth over the 10 years, hence I set 2015 as a base index of 100. I then utilised this base index for CPIH taking away each year from its previous to gather annual inflation data, private rent and annual earnings analysis. Additionally to calculate real earnings, I divided the Nominal Earnings Index by the CPIH index to gather a holistic conclusion on whether the UK was able to withstand the rising cost of living. Through gathering CPIH, Earnings and Private rent data, in separate Junyper notebooks, I converted each file to a csv file and exported all csv files into one master notebook. Finally, I used the library matplotlib to create a line chart showcasing all three values, comparing their increase over the decade.

Introduction:

The rising cost of living crisis has been a term popularised in the UK since 2021 creating a narrative that prior to this event, prices were relatively affordable compared to present day. Through utilising CPIH inflation data regarding different product types, I was able to collate a summary of whether certain price categories of items have been harder to afford. Additionally, I analysed housing data, another category considered a 'crisis,' I therefore wanted to test if this claim was factual or just the product of rosy retrospection.

Results

Figure 1

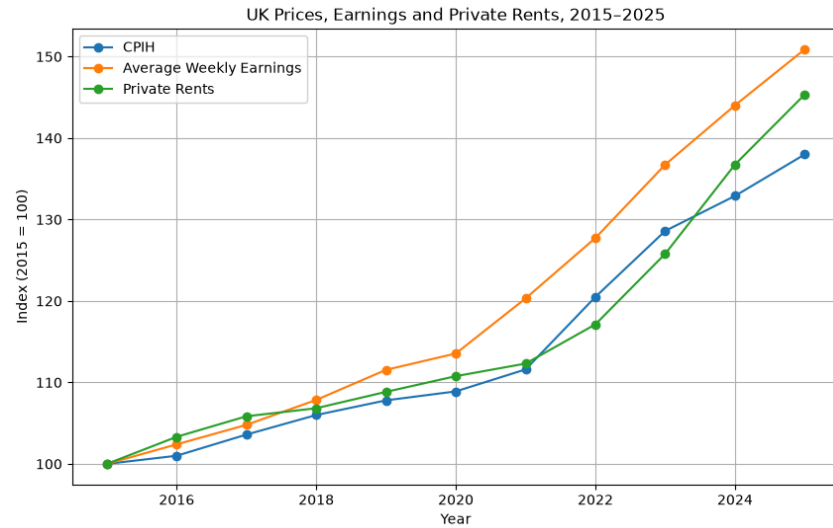


Figure 2

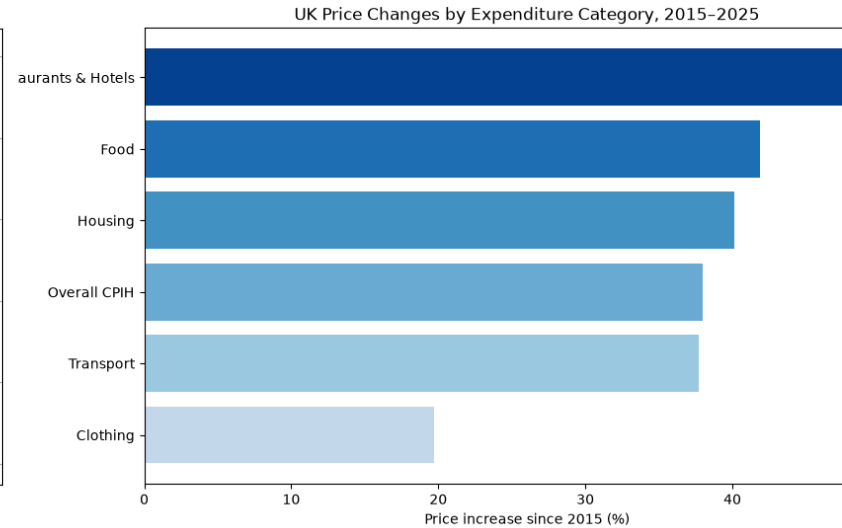


Figure 3

Year	CPIH	Inflation	Cumulative_change	Weekly_Earnings	Earnings_Growth	Earnings_Index	Real_Earnings_Index	Rent_Index_2015	Rent_Change	Earnings_minus_Inflation
2015	100.00	-	-	£481.98	-	100.00	100.00	100.00	-	-
2016	101.00	1.00%	1.00	£493.53	2.40%	102.40	101.38	103.33	3.33%	1.40
2017	103.60	2.57%	3.60	£505.08	2.34%	104.79	101.15	105.85	5.85%	-0.23
2018	106.00	2.32%	6.00	£519.84	2.92%	107.85	101.75	106.84	6.84%	0.60
2019	107.80	1.70%	7.80	£537.63	3.42%	111.55	103.48	108.84	8.84%	1.72
2020	108.90	1.02%	8.90	£547.30	1.80%	113.55	104.27	110.77	10.77%	0.78
2021	111.60	2.48%	11.60	£579.91	5.96%	120.32	107.81	112.32	12.32%	3.48
2022	120.50	7.97%	20.50	£615.66	6.16%	127.74	106.01	117.13	17.13%	-1.81
2023	128.60	6.72%	28.60	£659.06	7.05%	136.74	106.33	125.79	25.79%	0.33
2024	132.90	3.34%	32.90	£694.16	5.33%	144.02	108.37	136.75	36.75%	1.99
2025	138.00	3.84%	38.00	£727.31	4.77%	150.90	109.35	145.36	45.36%	0.93

Interpretation:

For most of the decade, earnings growth exceeded inflation suggesting positive real earnings growth. The Real Earnings Index increased from 100,0 in 2015 to 109.35 in 2025 suggesting purchasing power increased by 9.35% over 10 years, Although nominal earnings increased by 50.9% and CPIH by 38% suggesting an increase of 12.9%, the difference between the two values is solely just a difference between growth rates not an increase of purchasing power, since the real earnings index is more accurate, 9.35.

However for 2 years, inflation exceeded earnings growth: 2017 and 2022, having negative values in the earnings minus inflation column, -0.23 and -1.81. Research by Breinlich et al suggests in 2017 the depreciation of sterling following Brexit inflation increased import prices (Breinlich et al., 2017), whereas in 2022 the deterioration reflected a period of exceptionally high inflation with energy and food price shocks displacing real wage increases, despite weekly earnings growing 6.16% that year (Haskel et al., 2023). Following 2022, the economy was able to recover with 0.33, 1.99, and 0.93, values in the three subsequent years.

2020 experienced low amounts of inflation, 1.02 coinciding with the stagnation of worldwide economies due to Covid-19 through weaker consumer demand and substantial falls in oil and transport prices. Similarly 2016 experienced the lowest inflation partly reflecting the earlier decline in global oil prices creating reduced energy and transport costs too.

Figure 2 illustrates a divergence between essential and recreational goods with cost of living staples like food rising approximately 42% and housing, 40%, these being inelastic goods that consumers have to purchase regardless. Conversely, clothing experienced the slowest gain, 20% partly the product of globalised supply chains and fast fashion manufacturing efficiency. Transport being at approximately 37% reflects the decade of volatile fuel costs and changing post pandemic commuting patterns. Finally, restaurants and hotels increased by nearly 50%, perhaps the product of “revenge spending” (Simpson, 2024) where an influx of savings boosted spending on recreational services, leading to labour shortages and soaring energy bills for venues.

Rent prices increased by approximately 45.4% between 2015 and 2025 still lower compared to the 50.9% increase in earnings. This suggests that rents increased somewhat slower than the average earnings over the period. However, this data provides a general view of the entire country as there are substantial differences across locations, household sizes and incomes.

Conclusion

Overall, the given data does suggest that average earnings grew faster than CPIH producing higher real earnings in 2025 than 2015, Yet it does not conclusively prove that UK households were universally better off or increased by 9.35%. Living standards depend on factors beyond average wages and CPIH, including taxation affecting disposable income, household composition, distribution of income. The results therefore provide evidence of an improvement in purchasing power of average earnings rather than a complete measure of changes in living standards.

Bibliography

Simpson, J. (2024). UK attractions try to win back visitors as post-Covid 'revenge spending' ends. *The Guardian*. [online] 16 June. Available at: <https://www.theguardian.com/business/article/2024/jun/16/uk-attractions-try-to-win-back-visitors-as-post-covid-revenge-spending-ends> [Accessed 17 Sept. 2026].

Breinlich, H., Leromain, E., Novy, D. and Sampson, T. (2017). *The Consequences of the Brexit Vote for UK Inflation and Living Standards: First Evidence*. [online] Available at: https://personal.lse.ac.uk/SAMPSON/T/VoteInflation_TP.pdf [Accessed 17 Sept. 2026].

Haskel, J., Martin, J., Brandt, L., Albuquerque, D., Thomas, R., Daniel, H., Key, T., Goralczyk, G., Jeanes, H. and Khan, Z. (2023). *Recent UK inflation: an application of the Bernanke-Blanchard model*. [online] Available at: <https://www.bankofengland.co.uk/-/media/boe/files/speech/2023/november/recent-uk-inflation-an-application-of-the-bernanke-blanchard-model-paper.pdf> [Accessed 17 Sept. 2026].